

Past paper 2018

**Physical Chemistry
samester 6th**

**Short
Questions**



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. 012979

PAPER: Physical Chemistry

Course Code: CHEM-313 Part - II

TIME ALLOWED: 2 Hrs. & 45 Mins

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q. 2 Answers the following short questions:

(2×10=20)

- Define the term dynamic equilibrium.
- What is cage effect?
- Reaction in solution phase are faster than reaction in gas phase. Comment.
- Give a mathematical relation between standard Gibbs energy and equilibrium constant.
- What is the effect of temperature on vertical distribution of gas?
- Describe partition function.
- Write two Maxwell relations.
- What is difference between spontaneous and non-spontaneous process?
- Give two applications of Barometric formula.
- What do you understand by ionic strength?

Q.3 Derive Maxwell's law of distribution of molecular velocities **pg#15-20** (10)

Q.4 Describe Transition State theory. Derive Eyring equation to explain the effect of temperature on rate of reaction in solution. **pg#443-446** (10)

Q.5 (a) Discuss applications of partition function for evaluation of thermodynamic parameters. **pg#278-284** (5)

(b) Derive two Maxwell relations and give their significance. **pg#227-230** (5)

PHYSICAL CHEMISTRY

ASSIGNMENT

(041664)
PAST PAPER (2018)

QUESTION NO 1

SHORT Questions:-

(a)

DYNAMIC EQUILIBRIUM

A dynamic equilibrium is defined as.

The state of a reversible reaction in which the rate of forward reaction equals the rate of backward reaction and the concentrations of reactants and products remains the same.

Example:-

The reaction, $\text{NaCl} \rightleftharpoons \text{Na}^+ + \text{Cl}^-$ will be in dynamic equilibrium when the rate of dissolution of NaCl equals the rate of re-crystallization.

(b)

CAGE EFFECT

The reactions in solution are very fast as compared to those in gas phase due to **solvent cage effect**:-

"In cage effect "Molecules of different reactants are in a cage of the molecules of solvent and collide with each other many times and are converted into products in very short interval of time"

It describes how the properties of a molecule are affected by its surroundings.

(c)

Reactions in solution Phase are faster.

The reactions in solution phase

are very fast as compared to those in gas phase due to **Solvent cage effect**:-

In SOLUTION Phase:-

Molecules of different reactants are in a cage of the molecules of solvent and collide with each other many times and are converted into products in very short interval of time

In Gas Phase:-

In gas phase the probability of collision of the molecules of reactant is less as compared to that in solution.

(d)

Relation between Standard GIBBS Free energy and Equilibrium Constant

$$F = F^{\circ} + nRT \ln P$$

Free energy of a moles of A = $a \times F_A$

$$= aF^{\circ} + aRT \ln P_A$$

Free energy of b moles of B = $b \times F_B$

$$= bF^{\circ} + bRT \ln P_B$$

$$\text{Free energy of } c \text{ moles of } C = c \times F_c$$

$$= cF^\circ + cRT \ln P_c$$

$$\text{Free energy of } d \text{ moles of } D = d \times F_D$$

$$= dF^\circ + dRT \ln P_D$$

The change in free energy can be written as.

$$\Delta F = \sum n_i F_p - \sum n_i F_R$$

$$\Delta F = (cF_c + dF_D) - (aF_A + bF_B)$$

$$\Delta F = (cF^\circ + cRT \ln P_c + dF^\circ + dRT \ln P_D) - (aF^\circ + aRT \ln P_A + bF^\circ + bRT \ln P_B)$$

$$\Delta F = (cF^\circ + dF^\circ - aF^\circ - bF^\circ) + (cRT \ln P_c + dRT \ln P_D - aRT \ln P_A - bRT \ln P_B)$$

$$\Delta F = \Delta F^\circ + \{ RT \ln (P_c)^c + RT \ln (P_D)^d - RT \ln (P_A)^a - RT \ln (P_B)^b \}$$

$$\Delta F = \Delta F^\circ + \{ RT \ln (P_c)^c (P_D)^d - RT \ln (P_A)^a (P_B)^b \}$$

$$\Delta F = \Delta F^\circ + RT \ln \frac{(P_c)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

The quantity within logarithm will be equal to equilibrium constant.

$$0 = \Delta F^\circ + RT \ln \left[\frac{(P_c)^c (P_D)^d}{(P_A)^a (P_B)^b} \right]$$

$$0 = \Delta F^\circ + RT \ln K$$

$$\Delta F^\circ = -RT \ln K$$

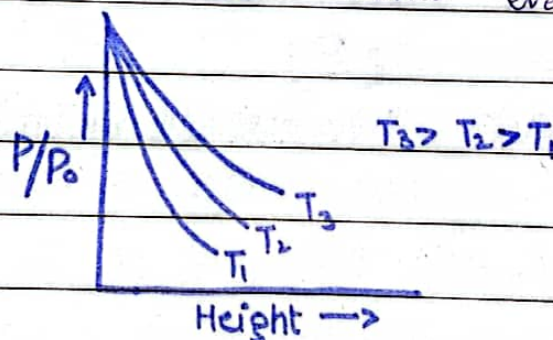
(e)
**EFFECT OF Temperature
ON Vertical distribution
OF Particles.**

According to equation.

$$P = P_0 e^{-Mgh/RT}$$

The value of P/P_0 at a given height will be larger at higher temperature than that at a lower temperature.

Thus, the variation in pressure with height is less gradual at higher temperature. If the temperature were infinite, the pressure would be same everywhere.



(f)
PARTITION FUNCTION

In quantum mechanics, all the information about microscopic system like position, velocity, momentum

energy is stored in a wave function Ψ .

In statistical mechanics also, there exists a function which contains all information about macroscopic system that function is known as **Partition function**.

It is a **dimensionless** quantity.

The partition function provides a bridge to link the microscopic properties of individual molecules with macroscopic properties of a system containing a large number of molecules.

It is denoted by symbol Q .

(8)

TWO Maxwell relations

The Maxwell relations consist of various characteristic functions, these are H , F , U , G .

It also includes thermodynamic parameters.

Pressure P , entropy S , volume V , and T .

Internal energy Maxwell Relation

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$$

Enthalpy Maxwell relation

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

(h) Spontaneous

- Spontaneous reactions take place on their own under a given set of conditions.
- The change in Gibbs free energy is **negative** for spontaneous reactions.
- Spontaneous reactions are **exergonic**.
- Increase entropy

Example:-

Combustion reactions

NON-Spontaneous

- Energy should be provided for the non-spontaneous reactions to proceed.
- In non-spontaneous the change in Gibbs free energy is **positive**.
- Non-spontaneous reactions are **endergonic**.
- Decrease entropy

Example:-

Reaction between atmospheric nitrogen and oxygen.

(i)

Applications of Barometric formula

Repeated

J IONIC STRENGTH

Definition:-

It is defined as half the sum of the terms obtained by multiplying the molality of each ionic species present in the solution by the square of its valence.

Ionic strength is given by symbol μ

Mathematically:-

$$\mu = \frac{1}{2} \sum m_i z_i^2$$

Where m_i is the molality of ion and z_i is its ionic charge. The product $m_i z_i^2$ is summed for all the ions in the solution then divided by two to give the ionic strength.